

AARON SUN

Experience

☎ 416-558-0210 ✉ a74sun@uwaterloo.ca 🌐 [linkedin.com/aaron-sun](https://www.linkedin.com/aaron-sun)

Plan Group

May 2026 – August 2026

Electrical Designer

Vaughan, Ontario

- Developed and modified **electrical building designs in Revit**, working within the power-design portion of project teams to produce coordinated electrical layouts and documentation.
- Created and updated **power distribution drawings and electrical layouts**, incorporating equipment, branch circuits, and distribution components into building designs.
- Coordinated electrical designs within Revit to maintain accurate **equipment locations, circuit layouts, and drawing documentation** across project revisions.
- Reviewed electrical drawings and design changes with the project team, gaining experience working within **engineering constraints, design standards, and multidisciplinary projects**.

Avnan

September 2025 – December 2025

Electronics Engineering Technician

Mississauga, Ontario

- Performed hands-on **PCB rework and repair** on production and prototype boards, including SMD and through-hole soldering, component replacement, and inspection under magnification.
- Conducted **functional and continuity testing** on assembled electronic products to identify manufacturing defects and verify circuit integrity.
- Supported the hardware team in **manufacturing validation and quality assurance**, gaining practical experience with electronics assembly and failure modes.

New Radio Tech Inc.

January 2025 – April 2025

Assistant Engineer

Shenzhen, Guangdong

- Refined automated **Python test scripts**, modularizing functions and adding error handling to reduce average test cycle time by **18%**.
- Performed bench-level electrical testing on **50+ prototype microchips**, using oscilloscopes, power supplies, and custom test fixtures to validate performance and isolate faults.
- Analyzed high-volume semiconductor test data using **Python, Pandas, and NumPy**, generating yield reports and trend visualizations to support root-cause analysis.
- Collaborated with design, firmware, and manufacturing teams to improve testing procedures and document standardized test processes.

Projects

RISC-V CPU | *SystemVerilog, Python, Yosys, Verilator*

- Implemented a **single-cycle RV32I CPU** in SystemVerilog, including the ALU, register file, control decoder, immediate generator, and memory subsystem.
- Built module unit tests, a **24-check full-core instruction regression**, and randomized differential tests against a Python reference model; passed **10,000+ randomized ALU instructions**.
- Developed a **two-pass Python assembler** with label resolution to generate test programs and automate branch and jump offset calculation.
- Ran **Verilator lint checks** and synthesized the RTL with **Yosys**, completing synthesis without errors.

STM32 MPU-6050 Breakout Board PCB | *Altium Designer, STM32, I2C*

- Designed a **4-layer PCB in Altium Designer** integrating an STM32F411 microcontroller and MPU-6050 accelerometer/gyroscope, completing schematic capture, component placement, and PCB routing.
- Integrated **3.3 V power regulation, decoupling capacitors, and ground planes**, with I2C sensor connectivity, a micro-USB connector, and an SWD programming/debug interface.
- Designed the **STM32F411 external clock circuit** using a **24 MHz crystal and load capacitors**, connecting it to the microcontroller's high-speed oscillator pins.

Technical Skills

Hardware: PCB Design, Schematic Capture, PCB Layout, SMD & Through-Hole Soldering, STM32, Arduino

EDA / Design: Altium Designer, KiCad, Revit, AutoCAD, ERC/DRC

Programming: C/C++, Python (Pandas, NumPy, Matplotlib), Verilog, SystemVerilog, VHDL

Tools: Git, GitHub, Linux, VS Code

Education

University of Waterloo

September 2024 – Present

Bachelor of Applied Science in Electrical Engineering, President's Scholarship of Distinction

Waterloo, Ontario